



Exponentials and Logarithms · Mark Scheme

Chapter 6: Exponentials and Logarithms. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Laws of logarithms

1 $\log_2 \left(\frac{9 \times 5}{15} \right)$ M1 = $\log_2 3$ A1.

2 (a) $4^{3/2} = 8$, so $\frac{3}{2}$ B1; (b) -3 B1.

3 $2\log_a x + 3\log_a y - \frac{1}{2}\log_a z$ M1 M1; $= 2p + 3q - \frac{1}{2}r$ A1.

4 $\log_6[x(x+5)] = 2$ M1; $x^2 + 5x - 36 = 0$ M1 A1; $(x+9)(x-4) = 0$; $x = 4$ ($x = -9$ rejected since $\log_6(-9)$ undefined) A1.

5(a) $\log_2 y - \log_2 x^2 = 3$ M1; $\frac{y}{x^2} = 2^3$ M1; $y = 8x^2$ A1.

5(b) $\log_3 \frac{x^2}{x-2} = 2$ M1; $x^2 = 9(x-2) \Rightarrow x^2 - 9x + 18 = 0$ A1; $x = 3$ or $x = 6$ A1; both satisfy $x > 2$, so both valid A1.

Section B – Equations

6 $x = \frac{\ln 20}{\ln 3}$ M1 = 2.73 A1.

7 $(x+1)\ln 2 = x\ln 5$ M1; $x(\ln 5 - \ln 2) = \ln 2$ M1; $x = \frac{\ln 2}{\ln 2.5} = 0.756$ A1.

8 $\ln \frac{2x+3}{x} = \ln 5$ M1; $2x+3 = 5x$ M1; $x = 1$ A1.

9 $y^2 - 6y + 8 = 0$ M1; $y = 2$ or 4 A1; $2^x = 2 \Rightarrow x = 1$; $2^x = 4 \Rightarrow x = 2$ M1 A1.

10(a) $(e^x - 2)(e^x - 3) = 0$ M1 A1; $x = \ln 2$, $x = \ln 3$ M1 A1.

10(b) $x\ln 2 + (2x+1)\ln 3 = \ln 15$ M1; $x(\ln 2 + 2\ln 3) = \ln 15 - \ln 3$ M1; $x\ln 18 = \ln 5$ A1; $x = \frac{\ln 5}{\ln 18}$, so $a = 5$, $b = 18$ A1.

Section C – Graphs

11 (a) $(0, 3)$ B1; (b) $y = 2$ B1.

12 (a) $x > -4$ B1; (b) $x = -4$ B1; (c) $(-3, 0)$ B1.

13 Correct shape (increasing, above asymptote) B1; $(0, -3)$ and $(\ln \frac{5}{2}, 0)$ B1; asymptote $y = -5$ B1.

14 $e^{2x} = 4e^{-x} \Rightarrow e^{3x} = 4$ M1; $x = \frac{1}{3}\ln 4$ A1; $y = e^{\frac{2}{3}\ln 4} = 4^{\frac{2}{3}} = 2\sqrt[3]{2}$ M1 A1.

15(a) Increasing ln-shaped curve with asymptote $x = \frac{3}{2}$ B1; crosses x -axis at $(2, 0)$ B1.

15(b) $y = \ln(2x-3) \Rightarrow e^y = 2x-3$ M1; $f^{-1}(x) = \frac{e^x+3}{2}$ A1; domain $x \in \mathbb{R}$ B1.

15(c) $\ln(2e^x - 3) = \ln 5 \Rightarrow e^x = 4$ M1; $x = \ln 4 = 2\ln 2$ A1.

Section D – Models

16 (a) 500 B1; (b) $500e^1 = 1359$ B1.

17 (a) £10976 B1. (b) $e^{-0.15t} = 0.25 \Rightarrow t = \frac{\ln 4}{0.15} = 9.24$ M1; during the 10th year A1.

18 (a) $\frac{1}{2}M_0 = M_0e^{-8k}$ M1; $e^{8k} = 2 \Rightarrow k = \frac{\ln 2}{8}$ A1. (b) $e^{-30k} = 2^{-3.75} = 0.0743$ M1; 7.43% A1.



19 (a) $A = 2000$ B1; $3500 = 2000e^{3k} \Rightarrow k = \frac{1}{3} \ln 1.75 = 0.187$ M1 A1. (b) $P(10) \approx 12900$ A1.

20(a) 95°C B1.

20(b) $60 = 20 + 75e^{-5k}$ M1; $e^{-5k} = \frac{40}{75} = \frac{8}{15}$ A1; $-5k = \ln \frac{8}{15} \Rightarrow k = \frac{1}{5} \ln \frac{15}{8}$ A1.

20(c) $10 = 75e^{-kt} \Rightarrow t = \frac{\ln 7.5}{k}$ M1 A1; $t = 16.0$ minutes A1.

20(d) $T \rightarrow 20^\circ\text{C}$ B1; e.g. the room temperature may not stay constant B1.

Section E – Linear form

21 $\log y = \log a + n \log x$ M1; $n = 3$ A1; $a = 5$ A1.

22 $\ln y = \ln a + x \ln b$ M1; $a = e^{1.2} = 3.32$ A1; $b = e^{0.4} = 1.49$ A1.

23 $\log y = \log 4 + \log 3^x = \log 4 + x \log 3$ M1 A1; gradient $\log 3$, intercept $\log 4$ B1.

24 $n = \frac{1.877 - 1.083}{0.699 - 0.301} = 1.99 \dots \approx 2$ M1 A1; $\log a = 1.083 - 2(0.301) = 0.481 \Rightarrow a = 3.03 \approx 3$ M1; $y = 3x^2$ A1.

25(a) Take logs of both sides and use product and power laws M1; form $Y = c + mt$ with constant gradient $\log q$ and intercept $\log p$ A1.

25(b) $\log p = 3.2 \Rightarrow p = 1580$ M1 A1; gradient $0.07 \Rightarrow q = 10^{0.07} = 1.17$ A1.

25(c) The value increases by about 17% each year B1 B1.

25(d) $t = 25$: $\log V = 3.2 + 25(0.07) = 4.95$ M1; $V \approx £89\,000$ A1.

Roar-level challenge

26 $\log_4(x+6) = \frac{\log_2(x+6)}{\log_2 4} = \frac{1}{2} \log_2(x+6)$ M1; $2 \log_2 x = \log_2(x+6) \Rightarrow x^2 = x+6$ M1; $(x-3)(x+2) = 0$ A1; $x = 3$ ($x = -2$ rejected since $\log_2(-2)$ is undefined) A1.